

18th Mexican Conference on Pattern Recognition

MCPR2026

June 24–27, 2026, CIITA, Ciudad Juárez, Mexico

Program

Wednesday, June 24 (Mexican Central Time UTC-5)	
9:00 – 9:20	Opening Ceremony
9:20 – 10:20	Keynote Address Chair: Prof. Humberto de Jesús Ochoa Domínguez <i>New Paradigms in Sensors and Computer Vision Methods + Comments on Artificial Intelligence Production at an International Scale</i> Prof. Walterio Mayol-Cuevas (Department of Computer Science, University of Bristol, United Kingdom)
10:20 – 10:40	COFFEE BREAK
	Session 1: Pattern Recognition and Machine Learning Techniques Chair: Prof. Erik Rodner
10:40 – 11:00	Evaluation of Clustering and Consensus Models for Classifying Adulterant Levels in Alcoholic Beverages Using Single-Shot Interferograms E. López-Meléndez, R. I. Álvarez-Tamayo, A. Barcelata-Pinzón, P. Prieto-Cortés, and L. D. Lara-Rodríguez
11:00 – 11:20	Beyond Pareto: A High-Efficiency Approach to Bi-Objective Regression Trees Erick G.G. de Paz, Arturo Hernández-Aguirre, and Iván Cruz-Aceves
11:20 – 11:40	Ensembles of Manifold Learners for Supervised and Unsupervised Learning Fabiola Muñoz Vera, Juan Manuel Pérez Ortega, Eduardo F. Morales, and Hugo Jair Escalante
11:40 – 12:00	Application of Unsupervised Pattern Recognition for the Structural Characterization of Masonry Walls Miguel Ángel Tlatzimatzi Flores, María Guadalupe Medina Barrera, and Eduardo Ismael Hernández
12:00 – 12:20	COFFEE BREAK
	Session 2: Pattern Recognition and Machine Learning Techniques Chair: Prof. Humberto de Jesús Ochoa Domínguez
12:20 – 12:40	Clustering Algorithms for Density Variations: An Empirical Comparison Adrián J. Ramírez-Díaz, José Fco. Martínez-Trinidad, and J. Ariel Carrasco-Ochoa
12:40 – 13:00	Dissimilarity-Based Graph Embedding via Local and Global Approximate Graph Patterns Daybelis Jaramillo-Olivares, J. Ariel Carrasco-Ochoa, and José Fco. Martínez-Trinidad
13:00 – 13:20	Comprehensive Validation of Brex Sequential Classifier Model for Robust Pattern Recognition Gerardo Acevedo-Sánchez, J. Pacheco-Senard, A. Alarcón-Paredes, Ó. Camacho-Nieto, and C. Yáñez-Márquez
13:20 – 13:40	A Machine Learning Framework for High-Performance Electrolyte Classification in Lithium Metal Batteries Sergio Rubén Ocampo-Pérez, Nouredine Lakouari, and Outmane Oubram
13:40 – 14:00	Dynamic Assessment of Landslide Susceptibility at Pico de Orizaba Using Ensemble Classifiers and Multitemporal Analysis Adhara Alejandra Avendaño Barajas, L. Altamirano Robles, R. Díaz Hernández, and S. Zapotecas Martínez
14:00 – 15:00	LUNCH
	Session 3: Deep Learning & Neural Networks Chair: Prof. Juan Humberto Sossa Azuela
15:00 – 15:20	Banking Fraud Detection Using Neural Network Models Ostin Uriel Martínez Campos, Andrés Ferreyra-Ramírez, Eduardo Rodríguez-Martínez, and Carlos Avilés-Cruz
15:20 – 15:40	Variational Standard Simplex Hierarchical Encoder for Clustering Ulises Rodríguez-Domínguez
15:40 – 16:00	Chest CT Gender Domain Translation: Comparative Evaluation of CycleGAN and cVAE in Radiometric and Volumetric Coherence I. Delgado Navarro, G. J. Alvarado Flores, L. Altamirano Robles, R. Díaz Hernández, and S. Zapotecas Martínez
16:00 – 16:20	Detection of Dangerous Wildlife in a Dynamic Environment Using Deep Learning David Cruz-Villavicencio and Jesús Alberto Martínez-Castro
16:20 – 16:40	COFFEE BREAK
	Session 4: Deep Learning & Neural Networks/Pattern Recognition and Machine Learning Techniques Chair: Prof. Walterio Mayol
16:40 – 17:00	Spatiotemporal Prediction of Wildfires Spread Using Deep Learning Ángel Itzcoatl Huizar Bretado, Raquel Díaz Hernández, K. A. Ramírez Gutiérrez, and L. Altamirano Robles
17:00 – 17:20	A Safety-Aware Approach for Automated Urine Culture Screening Using Calibrated Deep Learning G. J. Alvarado Flores, I. Delgado Navarro, R. Díaz Hernández, L. Altamirano Robles, and S. Zapotecas Martínez
17:20 – 17:40	Modeling Tumor Progression in Mammography with Conditional Generative Adversarial Networks Alfonso Rojas-Domínguez and Jesús Yaljá Montiel Pérez
17:40 – 18:00	Customer Preferences Recognition in Neuromarketing: Improving Accuracy with Autoregressive Modeling, Genetic Algorithms, and Naive Bayes Christian Ruiz-Ugalde, R. A. García-Hernández, J. Rojas-Simón, Y. L., and Marco Antonio Ramos-Corchado

18th Mexican Conference on Pattern Recognition

MCPR2026

June 24–27, 2026, CIITA, Ciudad Juárez, Mexico

Program

Thursday, June 25 (Mexican Central Time UTC-5)	
9:00 – 10:00	Panel in Spanish Chair: Prof. Osslán Vergara <i>Las Voces del Mañana: Niños Genio Debaten sobre Inteligencia Artificial</i> David Da Vinci y Luis Villareal
10:00 – 10:20	COFFEE BREAK
	Session 5: Medical Applications of Pattern Recognition Chair: Prof. Salvador Venegas
10:20 – 10:40	Machine Learning Algorithms for Identifying Attention-Deficit/Hyperactivity Disorder Processing Electroencephalographic Signals Erika Altair Castro-Verazas, Saúl Brandon Lima-Portillo, Delia Irazú Hernández-Farías, and C. A. Reyes-García
10:40 – 11:00	Graph-Based Modeling of Disfluent Speech for Alzheimer’s Detection Carlos A. Olachea-Hernández, Luis Villaseñor-Pineda, Manuel Montes-y-Gómez, and F. J. Martínez-Santiago
11:00 – 11:20	Factors Associated with Depression in Mexico: An XAI Comparative Analysis of Tree-Based Classifiers Under Imbalanced Data Abigail Romero-Trejo, A. Guzmán-Ponce, R.M. Valdovinos-Rosas, L. Cleofas-Sánchez, and I. Francisco-Valencia
11:20 – 11:40	COFFEE BREAK
	Session 6: Medical Applications of Pattern Recognition / Computer Vision Chair: Prof. Salvador Venegas
11:40 – 12:00	Hierarchical Incremental Learning for Adaptive Breast Thermography Classification Yareli Aburto-Sánchez, Pilar Gómez-Gil, and Leopoldo Altamirano-Robles
12:00 – 12:20	Automatic Classification of Subcutaneous Mycoses in Clinical Images Using Transfer Learning Vania Déborah Vázquez Palacios, Julio César Pérez Sansalvador, and Humberto Pérez Espinoza
12:20 – 12:40	Upsampling of Sparse Three-Dimensional Point Clouds Using iFactor-KDtree with Adaptive Interpolation Dora-Luz Almanza-Ojeda, Mario-Alberto Ibarra-Manzano, Carlos A. Pérez-Ramírez, and Yair A. Andrade-Ambriz
12:40 – 13:00	COFFEE BREAK
13:00 – 14:00	Keynote Address Chair: Prof. Juan Humberto Sossa Azuela <i>Foundation Models for Microscopy and Medical Data Analysis</i> Prof. Erik Rodner (Department of Data Science and Machine Learning, University of Applied Sciences, Berlin, Germany)
14:00 – 15:00	LUNCH
	Session 7: Language Processing and Recognition Chair: Prof. Ramón Iván Barraza Castillo
15:00 – 15:20	A Two-Stage Textual Preprocessing Pipeline for Medical Image Captioning S. Rascón-Cervantes, G. Ramírez-Alonso, A. Pastor López-Monroy, R. López-Santillán, and N. A. Rendón Mejía
15:20 – 15:40	Semantic Pattern Recognition in Scientific Literature Using Hybrid Topic Modeling and SciBERT With LoRA-Based Validation Yessenia Díaz Álvarez, Raúl Pinto Elías, A. Magadán Salazar, Noé A. Castro Sánchez, and J. Fuentes Pacheco
15:40 – 16:00	Automatic Depressive Symptom Detection on Social Media Using the BDI-II Cielo Aholiva Higuera-Gutiérrez, I. H. López-Nava, M. Montes-y-Gómez, M. Ezra Aragón, and D. E. Losada
16:00 – 16:20	Statistical Analysis of Combined Adversarial Attacks on Spam Detection Models Samantha Acosta Ruiz, Mireya Tovar Vidal, and José A. Reyes-Ortiz
16:20 – 16:40	COFFEE BREAK
	Session 8: Language Processing and Recognition / Computer Vision Chair: Prof. Ramón Iván Barraza Castillo
16:40 – 17:00	InCvT: A Hybrid Inception-Convolutional Vision Transformer Architecture for Speech Emotion Recognition Juan A. Ramírez-Quintana, Eduardo Gallegos-Camarena, A. A. Torres-García, and Verónica Gallegos-Orozco
17:20 – 17:40	Comparative Statistical Analysis of Sentiment Models in Spanish Political Discourse on YouTube Guillermo David Barrera Ortega, María Beatriz Bernabe Loranca, David Pinto Avendaño, and A. Carrillo Canán
17:40 – 18:00	Var Based on Knowledge Transfer from VLMs and Video Descriptions Emilio Vera-Cordero, David Mata Mendoza, Gibran Benitez-García, Hiroki Takahashi, and Mariko Nakano
19:00 – 23:00	CONFERENCE DINNER (Salón ARGOS del Centro de Convenciones Cibeles)

18th Mexican Conference on Pattern Recognition

MCPR2026

June 24–27, 2026, CIITA, Ciudad Juárez, Mexico

Program

Friday, June 26 (Mexican Central Time UTC-5)	
9:00 – 10:00	Keynote Address Chair: Prof. José Francisco Martínez Trinidad <i>Quantum Computing: Foundations, Transition to Industry, and Applications in Pattern Recognition</i> Prof. Salvador Elías Venegas Andraca (Tecnologico de Monterrey, Campus Ciudad de Mexico, MEXICO)
10:00 – 10:10	COFFEE BREAK
10:10 – 11:25	Panel in Spanish Chair: Prof. Juan Humberto Sossa Azuela <i>Tecnología con Propósito: Academia, Industria y Gobierno Dialogan sobre Inteligencia Artificial</i> Mtra. Elizabeth Burrola Meléndez (Centro de Innovación e Integración de Tecnologías Avanzadas) Mtro. Orlando Daniel Avitia (Director Regional Zona Norte de CECyTECH) Prof. Salvador Elías Venegas Andraca (Tecnologico de Monterrey, Campus Ciudad de Mexico) Ing. Javier Acosta (Mechatronics AI)
11:25 – 11:40	COFFEE BREAK
	Session 9: Computer Vision Chair: Prof. Jesús Ariel Carrasco Ochoa
11:40 – 12:00	Embedded System for Vehicle Environment Perception and License Plate Recognition (LPR) Using Computer Vision and Deep Learning R. Leonardo Méndez-Macías, J. Villegas-Cortez, A. Ferreyra Ramírez, A. Zúñiga-López, and S. Cordero-Sánchez
12:00 – 12:20	Interpretable Human Activity Recognition for Subtle Robbery Detection in Surveillance Videos Bryan Jhoan Cazáres Leyva, Ulises Gachuz Dávila, José Juan González Fonseca, Juan Irving Vásquez, Vanessa A. Camacho-Vázquez, and Sergio Isahí Garrido-Castañeda
12:20 – 12:40	Tree Crown Segmentation in UAV RGB Images via Monocular Depth Estimation Sergio Adán-Juárez, Andrea Magadán-Salazar, J. Fuentes-Pacheco, Raúl Pinto-Elías, and J. Tavira-Villanueva
12:40 – 13:00	UAV Pose Estimation in Low-Light Conditions Using Visual-LiDar Fusion Esteban Tlelo-Coyotecatl, Alejandro Gutiérrez-Giles, and José Martínez-Carranza
13:00 – 13:30	Closing Ceremony
13:30 – 15:00	LUNCH

18th Mexican Conference on Pattern Recognition

MCPR2026

June 24–27, 2026, CIITA, Ciudad Juárez, Mexico

Program

Saturday, June 27 (Mexican Central Time UTC-5)			
8:30 – 9:00	Registration		
9:00 – 11:30	Tutorial 1 <i>Understanding Modern Natural Language Processing Techniques: Tokenization, Embeddings, and Transformer Architectures</i> Prof. Humberto de Jesús Ochoa Domínguez Universidad Autónoma de Ciudad Juárez, Ciudad Juárez, MEXICO	Tutorial 2 <i>Neuromorphic Computing for Edge AI: A Practical Introduction with SPICE and Verilog-A</i> Prof. Victor Hugo Ponce Ponce Centro de Investigación en Computación del Instituto Politécnico Nacional, CDMX, MEXICO	Tutorial 3 <i>Robot Modeling and Control</i> Prof. Elsa Rubio Espino Centro de Investigación en Computación del Instituto Politécnico Nacional, CDMX, MEXICO
11:30 – 11:45	COFFEE BREAK		
11:45 – 13:30	Tutorial 1	Tutorial 2	Tutorial 3
13:30 – 14:50	LUNCH		